

First try with Yelp Fusion API: Here's a rough conclusion

A study of Blue Bottle Coffee & Joe Coffee Company

Post Link:

https://open.substack.com/pub/juliaqian/p/first-try-with-yelp-fusion-api-heres?r=4mh4cl&utm_campaign=post&utm_medium=web&showWelcomeOnShare=true

Casual thoughts from a busy coffee lover in NYC

Living in New York City means always being on the move, and as a coffee lover with little time to brew at home, I've come to rely on two favorite chains: **Blue Bottle Coffee** and **Joe Coffee Company**. Their coffee feels like a treat compared to Starbucks, which, while convenient, doesn't quite hit the same quality note for me. I realized recently that while these two brands feel like similar choices—both with clean, cozy vibes and a shared blue color scheme—their locations got me thinking:

- How are Joe Coffee and Blue Bottle distributed across Manhattan?
- Do these distribution patterns reflect the neighborhoods they target? My hunch is that their presence relates to factors like demographics or income levels.

Coffee chains often reveal more than they seem—they reflect local trends, the people living there, and how residents spend their time and money. Recent research backs this up, showing that businesses like coffee shops can potentially serve as **early indicators of gentrification**, hinting at changes in a neighborhood's identity and socioeconomic profile. Inspired by these ideas, I decided to explore these two brands more closely.

Why focus on coffee chains?

Usually ignored, but frequently proved by academic literature, coffee stores reflect trends and demographics: who lives in an area, what they value, and how they spend their time and money. Recent studies in the paper *Nowcasting Gentrification: Using Yelp Data to Quantify Neighborhood Change* explain the traits of gentrification using Yelp data, and show that businesses often act as leading indicators of neighborhood change. My study is therefore trying to mimic their analysis, of course in a much simpler way, by focusing on two coffee brands: Joe

Coffee Company—a neighborhood-friendly, community-based chain (as they describe themselves), and Blue Bottle Coffee—a store with sleek, upscale vibe and minimalist aesthetics.

Case Study Site Selection

Using the *Yelp Fusion API*, I gathered store data across Manhattan for both chains, focusing only on open, active locations with consistent reviews. Due to API limitations, I retrieved information on *11 Joe Coffee locations* and *19 Blue Bottle stores* from public records and research.

Manhattan was chosen because its diverse neighborhoods provide a snapshot of distinct urban identities, from community-driven enclaves to fast-paced, professional hubs.

Key Observations:

- **Joe Coffee Company** clusters around areas like the Upper West Side and Chelsea (zipcode: 10023, 10024, 10025, 10001, 10011, 10018), known for residential communities, schools, and parks. This may reflect they are very community-oriented.
- **Blue Bottle Coffee** is more prevalent in downtown neighborhoods, such as Soho, Tribeca, and the Financial District (zipcode: 10012, 10013, 10014, 10007, 10004, 10005, 10006, 10038). These may align with its image of luxury and minimalism, attracting professionals in higher-income brackets.
- Both may indicate their preference for high-income areas as location.

Finding & Analysis

*Here because this practice is about API, I didn't map out the specific location on the actual NYC map, the analysis is a mixture of data filtering and online manual research.

Based on the data I retrieved, I found the location & zip code of all Joe Coffee stores (22 stores data available) in areas like the Upper West Side and Chelsea, places known for their mix of residential, academic institutions, and community-based culture. Joe Coffee locations were also clustered near schools and parks, focusing on fostering a sense of community.

On the other hand, Blue Bottle Coffee had a distinct concentration in downtown neighborhoods—like Soho, the Financial District, and Tribeca—areas stereotypically considered as with higher income levels and a strong presence of professionals. These spots somehow align with Blue Bottle’s identity as a “luxury brand offering premium coffee experiences”.

Therefore, just based on the location selections, we may say Blue Bottle’s presence speaks to the demand for sleek, high-end coffee experiences in affluent zones, while Joe Coffee’s strategy seems more focused on accessibility and community connection, as also calls back to its brand setting. And standard-wise, the two stores do not have big differences in the Yelp platform rating system.

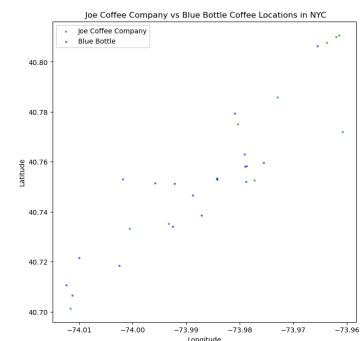
*Joe Coffee Company has a higher average rating than Blue Bottle Coffee.

*Blue Bottle Coffee Average Rating: 3.83

*Joe Coffee Company Average Rating: 3.96

(1) Clustering and Proximity Patterns: Are They Targeting the Same Crowd?

Interestingly, it seems like very few overlapping locations between Joe and Blue Bottle. When they did coexist, it was usually in neighborhoods with a healthy blend of professional residents and students—like the West Village(10013, 10014) or Morningside Heights(10025, 10027). This suggests that while their core audiences may overlap slightly, each brand has found a comfortable niche without significantly encroaching on the other’s territory.



(2) More of Coffee Chains and Neighborhood Identity

Each chain’s clustering tells a story about the identity of its neighborhoods. Joe Coffee’s focus on quieter, community-centric areas mirrors the brand's friendly vibe, emphasizing comfort and inclusiveness. Blue Bottle’s downtown strategy, by contrast, fits the high-paced

lifestyle of its customers, offering fast but premium service in spaces that mirror the aesthetics of modern minimalism.

*Indication of potential clustering (not actual position)

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price	Median_household_income
0	Blue Bottle Coffee	22 Broad St, New York, NY 10005	40.706510	-74.011230	10005	4.3	22	N/A	197188
1	Blue Bottle Coffee	396 Broadway, New York, NY 10013	40.718480	-74.002480	10013	4.1	37	\$\$	137572
2	Blue Bottle Coffee - New York	150 Greenwich St, New York, NY 10007	40.710715	-74.012330	10007	4.2	175	\$\$	250001
3	Blue Bottle Coffee	408 Greenwich St, New York, NY 10013	40.721594	-74.010010	10013	3.4	8	N/A	137572
4	Blue Bottle Coffee	101 University Pl, New York, NY 10003	40.734030	-73.992520	10003	4.1	62	\$	137533
5	Blue Bottle Coffee	1227 Broadway, New York, NY 10001	40.746621	-73.988791	10001	3.9	11	N/A	101409
6	Blue Bottle Coffee	441 8th Ave, Ste 68, New York, NY 10001	40.751537	-73.995835	10001	3.7	20	N/A	101409
7	Blue Bottle Coffee - Hudson Yards	20 Hudson Yards, Ste 228, New York, NY 10001	40.753119	-74.001823	10001	3.5	97	\$\$	101409
8	Blue Bottle Coffee - Bryant Park	54 W 40th St, New York, NY 10018	40.753064	-73.984223	10018	4.0	331	\$\$	130457
9	Blue Bottle Coffee	54 W 40th St, New York, NY 10018	40.753080	-73.984160	10018	3.7	9	N/A	130457
10	Blue Bottle Coffee	10 E 53rd St, New York, NY 10022	40.759618	-73.975489	10022	3.9	86	\$\$	156255
11	Blue Bottle Coffee - Amsterdam Ave	279 Amsterdam Ave, New York, NY 10023	40.779306	-73.980923	10023	3.8	56	\$\$\$	136109
12	Blue Bottle Coffee - Morningside Heights	2901 Broadway, New York, NY 10025	40.806390	-73.965420	10025	3.2	29	N/A	103440
13	Blue Bottle Coffee - Gramercy Park	257 Park Ave S, New York, NY 10010	40.738591	-73.987138	10010	3.6	44	N/A	131635

	Name	Address	Latitude	Longitude	Zip Code	Rating	Review Count	Price	Median_household_income
0	Joe Coffee Company	141 Waverly Pl, New York, NY 10014	40.733297	-74.000566	10014	4.1	380	\$	136559
1	Joe Coffee Company	9 E 13th St, New York, NY 10003	40.735171	-73.993262	10003	3.9	359	\$\$	137533
2	Joe Coffee Company	7 South St Slip, New York, NY 10004	40.701345	-74.011612	10004	3.4	18	N/A	216017
3	Joe Coffee Company	514 Columbus Ave, New York, NY 10024	40.785768	-73.972978	10024	4.0	396	\$	153177
4	Joe Coffee Company	187 Columbus Ave, New York, NY 10023	40.775080	-73.980367	10023	4.0	143	\$	136109
5	Joe Coffee Company	1045 Lexington Ave, New York, NY 10021	40.772044	-73.960805	10021	4.1	261	\$	136060
6	Joe Coffee Company	550 W 120th St, New York, NY 10027	40.810050	-73.962020	10027	3.8	216	\$\$	58435
7	Joe Coffee Company	55 W 40th St, Ste 9, Bryant Park, New York, NY...	40.753458	-73.984319	10018	3.3	26	\$\$	130457
8	Joe Coffee Company	105 E 42nd St Grand Central Dining Concourse, ...	40.752731	-73.977233	10017	4.1	9	N/A	136112
9	Joe Coffee Company	305 Dodge Hall, 305 Dodge Hall, Upper Campus L...	40.810686	-73.961453	10027	5.0	6	N/A	58435
10	Joe Coffee Company	2950 Broadway Pulitzer Hall, New York, NY 10027	40.807631	-73.963733	10027	3.9	8	N/A	58435

*Blue Bottle Coffee stores are located in areas with higher average household income.

*Blue Bottle Coffee Average Median Household Income: 139460.43

*Joe Coffee Company Average Median Household Income: 123393.55

The Limits of the Grind: Challenges and Considerations

This study comes with some limitations, for sure. First, while Yelp data gives us valuable insights, it reflects only businesses that are active on the platform, which means the results can be skewed, especially in areas where Yelp adoption is uneven (at least I personally always refer to Google Maps for store review). Additionally, the analysis focuses on Manhattan alone, which may exclude trends happening elsewhere in New York City.

Also, it’s worth noting that these clustering patterns are very rough, and the conclusion I tried to draw is based on my personal understanding & online description of the general situation of locations with certain zip codes. And of course, businesses often pivot to adapt to new consumer trends or respond to competitors, meaning today's insights may look very different next year.

Still, even with all the limitations, I do think the findings highlight how coffee chains are more than just places to grab a latte—they reflect the evolving dynamics of urban life. Joe Coffee’s or Blue Bottle’s may look the same to many, but these choices may also reflect a lot about how businesses respond to their environments, and in turn, how consumers shape those environments with their preferences and spending habits.

Next Step... ?

As Manhattan continues to evolve, it will be fascinating to watch how these brands grow—and whether new players emerge to shake up the scene. Will Joe Coffee expand into trendier areas, or will Blue Bottle try to capture a slice of the community-centric vibe? In any case, tracking these shifts provides a valuable lens for understanding the interplay between business strategy and neighborhood identity.

Though brief in my research, I hope in my future practice, the investigation won't be just about the coffee. I also want to explore about what these brands symbolize and how their presence aligns with local lifestyles, preferences, and affluence.

So, next time you walk past a Joe Coffee or grab a flat white at Blue Bottle, remember: That coffee cup is part of a larger story. It's not just about caffeine. It's about how cities evolve, how businesses adapt, and how neighborhoods define themselves through the choices we make every day—one cup of coffee at a time.